

Human-Centered Design for an In-Vehicle Truck Driver Fatigue and Distraction Warning System

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Abstract—Driver fatigue and distraction are major road safety issues globally; developing driver state detection and warning technology to help reduce impairment-related incidents is a promising approach. The aim of this case study was to design an effective Human Machine Interface (HMI) for a near-market driver warning system primarily aimed at commercial truck driving. A human-centered design (HCD) process was employed for the development and evaluation. Application of HCD here was a multi-stage iterative process: a comprehensive literature review; developing a context of use description; undertaking truck driver interviews; identifying user needs and associated design requirements; conducting two design workshops; operationalizing the design; running HMI evaluation studies, and finalizing the HMI concepts. As a result of the iterative HCD process, the HMI comprises a multi-modal warning system (visual, auditory and tactile) with two levels for driver fatigue and an escalating system for driver distraction. Following this successful HCD process, further on-road evaluation work is proposed before the final version of the HMI is manufactured.

Index Terms—Human centered design, driver fatigue, driver distraction, driver warning systems.

I. INTRODUCTION

GIVEN the vital importance of the road freight transport sector in maintaining global productivity and competitiveness, the safety and well-being of heavy vehicle drivers and other road users is of paramount importance [1]. This is particularly pertinent given the high mileage that truck drivers in Australia, North America and around the world routinely cover, and the fact that injury outcomes resulting from heavy vehicle collisions are typically more much severe than those associated with collisions involving other vehicles [2].

Both driver fatigue and distraction are major challenges in the commercial vehicle safety context [3]. Driver distraction can be defined as the diversion of attention away from activities critical for safe driving toward a competing activity [4].

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Driver fatigue can be defined as a state of reduced mental alertness: it is most commonly attributable to prolonged time-on-task and/or sleepiness which is driven by circadian rhythms and generally increases with the length of wakefulness. To address these issues, regulators and fleet operators often implement a range of measures designed to ameliorate crash risk and improve driver well-being; these include scheduling and allowable work hours, including limits on operational time, rest breaks and work diary requirements [5]. While these regulations are important, recent advances in vehicle technology permit the real-time monitoring of driver fatigue and driver distraction that has potential for further improved road safety outcomes [1], particularly if such technology is introduced as part of an integrated fatigue and distraction management plan. Such a plan may include a well-designed interface for the technology (so that warnings are seen/heard, understood and heeded), post-trip feedback and advice following critical events, driver education about the importance of being well-rested and frequent breaks during shift [5].

Many commercial vehicle manufacturers and after-market suppliers have developed, or are in the process of developing, in-vehicle technology that seeks to reduce the risks associated with fatigue and distraction (for example, [5], [6]). However, relatively little published research has evaluated the effectiveness of the HMI design for these systems. This is an important gap since a key issue affecting the efficacy of any in-vehicle technology to reduce fatigue and distraction is the user interface; including how information is transmitted to operators and its usability and acceptability among drivers. Within the context of driver monitoring technology, this includes defining the most effective warnings for heavy vehicle drivers and how they should be delivered given the driver's operational constraints, their driving task and other technologies present in a truck's cabin.

A. Human Centered Design

To develop an effective driver interface for an in-vehicle driver warning system, the use of a state-of-the-art design process that places at its core driver requirements is central. To this end, HCD focuses technology design on end-users taking into consideration the organizational and environmental contexts in which driving tasks are performed. The HCD process requires the involvement of end-users and other stakeholders throughout the entire design process [7]).

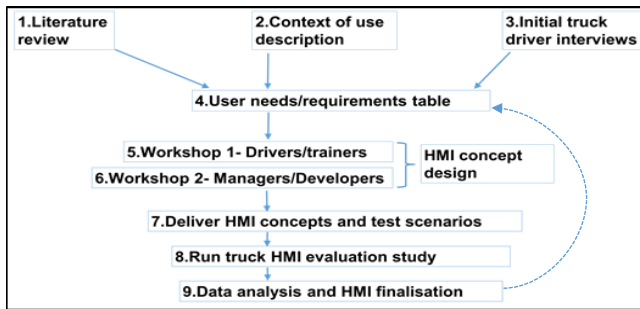


Fig. 1. HCD approach for in-vehicle driver warning system development.

A key strength of the HCD process is that it is iterative; that is, it is not a linear process where a single idea remains unchanged from concept through to production [8]. As such, the process, especially for complex automotive technology as described in this case study, usually involves revisiting, refining and changing design ideas, and then testing, evaluating and further refinement [7].

ISO 9241: 210 Human-Centered Design for Interactive Systems [8] provides a general framework for HCD activities that is applicable to this automotive case study; it specifies that four linked HCD activities should take place during the design of the system: Understand and specify the context of use, specify user requirements, produce designs to meet these requirements and evaluate the designs. Previous studies have applied a HCD-type process to the development of driver warning technology in road [9], [10] and air transport [11] but not specifically in relation to HMI design for driver fatigue or distraction. An exception is a 2003 study [12]; although it applied a user-centered design to the development of a driver fatigue warning system, the prototype is largely now outdated in view of the myriad in-vehicle safety and information systems currently operable in modern day vehicles.

To the authors' knowledge, a contemporary HCD process has not been applied to the design of in-vehicle driver warning technology aimed at mitigating driver fatigue and distraction events. This case study aimed to develop an effective HMI for a driver warning system using a HCD process.

II. METHODOLOGY

A human-centered design approach was used that built on the framework outlined in ISO 9241: 210 [8] and further informed by a specific application of HCD for new vehicle technology by Horberry *et al.* [7] and successful participatory ergonomics programs [13]. A comprehensive nine stage HCD strategy for the design and evaluation of the HMI for in-vehicle warning systems was deployed and is presented in Fig 1 and elaborated on in Table I.

The outcomes of these nine stages are described in the outcomes section below.

III. OUTCOMES

A. Literature Review

The literature review focused on four components:

- Research on the effectiveness of HMI design for fatigue and distraction warning;

TABLE I
DESCRIPTION OF STEPS IN THE HCD APPROACH FOR
IN-VEHICLE DRIVER WARNING SYSTEMS

Steps	Purpose and intent
1. Literature review	To identify research on HMI design found to be effective for fatigue and distraction warning and to review existing HMI specifications of in-vehicle systems.
2. Context of use description	Working with stakeholders, to describe what the in-vehicle driver warning system device should do, what functionality it should have, what information should be displayed, when and in which modalities.
3. Initial truck driver interviews	Interviews with a representative group of truck drivers to better understand user needs. This included drivers that had a range of experience with driver warning systems and other technologies.
4. User needs/requirements	Building on the first three stages presented above, user needs were uncovered. From this, design requirements to meet those needs were then identified.
5. Workshop 1- Drivers/trainers	To iteratively design the HMI based on the user needs/requirements and end-user input in a participatory design workshop with drivers and driver trainers.
6. Workshop 2- Managers and/or developers	To iteratively design the HMI, based on the user needs and other stakeholder input in a participatory design workshop with managers (with prior input from system developers).
7. Deliver HMI concepts	To operationalise the HMI design concepts (e.g. exact sizes, tones, positions for the warnings etc.) based on the emerging design.
8. Run truck HMI evaluation study	Given the ethical and safety concerns of evaluating a prototype system on real-roads the evaluation included both inspection-based and end-user based testing in a truck driving simulator.
9. Design Finalisation	To produce a detailed HMI design and to outline further development work, such as on-road testing.

- Research on the effectiveness of HMI design for collision warning under fatigue or distraction;
- General automotive HMI design guidelines relating to the safety of in-vehicle information and communication systems, and
- HMI designs of currently available fatigue and distraction warning systems.

1) *Research on HMI Effectiveness for Fatigue and Distraction Warnings:* There was relatively limited research on the effectiveness of fatigue or distraction warning HMI in the absence of an external road/traffic-based hazard or imminent collision.

Dangelmaier and Marberger [14] found that a multimodal warning comprising an auditory (ding or 3 beep sound) plus vibration through the seat belt and a visual alert was more effective than unimodal combinations in alerting task induced drowsy drivers. Similarly, Ayoob *et al.* [12] found a high level of acceptability among drivers for a prototype two-stage fatigue warning HMI comprising multimodal auditory and visual alerts: drivers also preferred multimodal (redundant) warnings over unimodal warnings especially for severe levels of drowsiness. A field operational test showed that drivers had lower PERCLOS values (percentage of eyelid closure over the pupil over time) compared to baseline, although there were

no differences in safety critical events between the conditions [15]. Drivers who rated the system favorably responded more quickly to a valid alert than drivers who rated it unfavorably and were generally more accepting of it.

Aidman *et al.* [16] used a single stage visual-auditory interface to warn drivers when their fatigue reached a medium-high risk range. The warning resulted in reduced fatigue and improved alertness and driving performance ratings. Fitzharris *et al.* [3] examined the effectiveness of a driver fatigue monitoring system which issued warnings via simultaneously presented auditory and haptic alerts. Compared to baseline, the combined warnings resulted in a 66% reduction in fatigue events.

Studies evaluating the effectiveness of driver distraction warning HMI used single-stage warnings comprising visual-auditory [17]; visual [18], [19] or tactile alerts [20]. In response to visual distraction these systems demonstrated safety benefits including improved braking or reduced time-to-collision [17]; increased on-road attention [18]–[20] or decreased glances to a portable in-vehicle device [18], [19].

2) *Research on the Effectiveness of HMI Design for Collision Warning Under Fatigue or Distraction:* A number of studies examined the effectiveness of collision warning systems for drivers under fatigued or distracted states. Baldwin *et al.* [21] found that an auditory collision warning presented as a non-verbal 1,000 Hz tone or a verbal warning ('danger!') reduced crashes following task induced fatigue. Gaspar *et al.* [22] found that two types of auditory forward collision warnings – an auditory alert (series of loud beeps) and a haptic seat vibration (series of rapid pulses) – were more effective than baseline in reducing response time and increasing safety margins in an overnight study for drivers who were looking down or had their eyes closed prior to the collision event.

Collision warning systems have consistently produced more effective responses compared to a no warning condition when drivers were distracted by visual [23]–[25], cognitive [25], [26]–[29] or combined visual and cognitive secondary tasks [30], [31]. These effects have been demonstrated for both single modality HMIs including auditory [23], [24], [26], [27], [31], [32], and tactile alerts [26], [27], [29], [32] as well as multi-modal HMIs including audio-visual alerts [25], [28], [30], auditory-haptic alerts [26], [27] or visual-haptic alerts [28].

Cross-modal comparisons of collision warning HMIs have generally shown that tactile alerts were more effective than auditory alerts under cognitive [14], [33] or visual distraction [34] and were perceived by drivers as being more effective and less annoying [28], [35]. Visual warnings were not effective for attracting the attention of drivers who were looking away [36] and were also ineffective, or not as effective, as warnings presented in the auditory or haptic modalities during urgent collision situations for attentive drivers [36]–[38].

Studies have also shown that multimodal warning presentations were more effective than unimodal warning presentations in reducing visual distraction [39], [40], [26] and cognitive distraction [26], [27], largely because redundancy provides an additional means of attracting the driver's attention [26],

[27], [36], [39]. Recent research conducted as part of the ADAS&ME project also found that drivers impaired by distraction or fatigue prefer multimodal warnings in the event that a single modality HMI is missed or inoperable [28], [41]. Despite these findings, it is important not to overload or annoy the driver from the display of redundant information when using simultaneously presented multimodal warnings.

Multi-stage collision warning systems (cautionary warning followed by an urgent warning) were found to be more effective in alerting cognitively distracted drivers than single stage collision warning systems (comprising only an urgent warning) [28]. However, under visual distraction, single stage warnings appeared to be more effective [34] or at least no different to multi-stage warnings [24]. The most effective multi-stage collision warning systems generally comprised a cautionary visual alert followed by a combined visual-auditory alert at the urgent warning stage [19], [36], [40], [42]. In some cases, drivers also preferred an escalating tactile warning for both fatigue and distraction warnings [35].

3) *General Automotive HMI Design Guidelines:* Fourteen sets of automotive HMI design guidelines relating to the safety of in-vehicle information and communication systems were reviewed, from which high-level principles were derived to inform the design of the driver warning HMI. Only one set of HMI guidelines specifically addressed fatigue warnings [43] and none of the guidelines specifically addressed distraction warnings.

The principles most relevant to this case study provided specific guidance on warning system design for collision avoidance under both urgent (i.e. imminent) and non-urgent (i.e., cautionary or advisory) situations. Key specifications included communicating urgency via increasing certain acoustic properties for auditory alerts, such as the frequency, intensity, and temporal rate of a signal [36], [43]–[47], and by changing the size, color and flash rate of the warning icon for visual alerts [36], [43], [44], [47].

The HMI guidelines also recommended that any warning system should not be able to be disabled by the driver or interfere with other warning systems or driving tasks [47]–[50]. They also specified that only minimal training should be required for system understanding [47], [49], [50] and false alarms should be reduced as much as possible to prevent driver annoyance and mistrust [36], [45].

4) *Current Driver Fatigue and Distraction Warning Systems:* The review identified 16 Original Equipment Manufacturer fitted warning systems. Of these, 10 were described as fatigue warning, and six were both fatigue and distraction or attention monitoring. There did not appear to be any publicly available evaluations of the HMI design concepts for these systems.

There was little difference between the HMIs that were described as fatigue only and those that were described as both fatigue and distraction warning systems. Most systems issued simultaneously presented multimodal warnings (visual and auditory), with the typical interface comprising an illuminated coffee cup icon together with a short text message shown on the vehicle display advising the driver to take a break. Few systems specified the level of impairment required to trigger

the system, the characteristics of the auditory warning/s (tone, pitch etc.), or the frequency with which warnings were issued. In a small number of fatigue and distraction warning systems the warnings comprised continuous visual feedback displayed via illuminated bars on the dashboard that faded as the level of attention decreased.

Some systems issued multi-stage warnings that escalated in urgency as fatigue level increased or distraction persisted. These typically comprised simultaneously presented audio-visual warnings at both the cautionary warning stage and the urgent pre-collision stage. In one of these systems, tactile alerts in the form of steering wheel vibrations together with audio-visual warnings were used to prompt urgent responding. One of the fatigue monitoring systems was able to be used in conjunction with longer term performance data for overall risk management; this included sending notifications to managers in real time regarding the driver's current fatigue level.

5) *Summary of Literature Review Findings:* Overall, the findings arising from the literature review for both fatigue and distraction warning included the use of a two-stage system comprising an initial lower-level cautionary warning followed by an urgent warning. For both states, it is concluded that the urgent warning should comprise multi-modal visual-auditory alerts and, potentially, tactile warnings presented either via steering wheel vibration or seat vibration. The cautionary stage should comprise a low-level multi-modal auditory-visual alert for fatigue and a low-level auditory warning for distraction.

Overall, the absence of guidelines and empirical literature specific to the development of driver fatigue or distraction warning HMI in the absence of an external hazard or imminent collision further underscores the importance of the HCD process in this case study.

B. Context of Use Description

In conjunction with the literature review, and working with industry end-users comprising a national trucking company and an international driver technology manufacturer, the following definition of the system's overall purpose was derived:

The overall goal of a fatigue and distraction warning system is to reduce fatigue and distraction related incidents and subsequent road trauma. The system should effectively detect fatigue and distraction when it occurs at a level that might impact upon safe driving performance in drivers. It should effectively display that information both to the driver in real time and record it for training or management purposes. The choice of display modalities is currently open, but it is likely that visual, auditory and tactile warnings may be used. Use of a warnings hierarchy is likely (e.g. first alerting/caution, then a more salient warning).

Based on this definition, and the findings of the literature review, initial design concepts for visual, auditory and tactile in-cabin alerts and warnings were created.

C. Initial Truck Driver Interviews

1) *Participants:* Sixteen truck drivers ranging in age from 24 to 65 were recruited from a large trucking company and from the general population of heavy vehicle drivers within

Victoria using on-line advertisements. All participants held a current heavy vehicle license and had been driving trucks in Australia for an average of 19.4 years (SD=14.5). Twelve drivers had an average of 18 months in-cab experience with driver warning systems. The remaining four drivers had no previous driver warning system experience.

2) *Materials:* Drivers were interviewed separately to determine their views on a description of the initial HMI concepts for fatigue and distraction warning. An interview protocol was developed to ensure consistency of content. The interview questions addressed the following key themes: the driver's previous experience with driver warning systems; perceived advantages and disadvantages of a proposed system; likely acceptance and perceived effectiveness of visual, auditory and tactile warnings; likely acceptance and perceived effectiveness of single versus multi-stage warning components; driver views regarding placement of the system within the cabin and opinions about driver training with respect to use of the system. Approval for the research was obtained by the Monash University Human Research Ethics Committee (Project Number 8247). A summary of key themes is presented below.

3) *Perspectives on Driver Warning Systems:* Overall, there was strong support from truck drivers for the initial HMI concepts, particularly among those with prior experience of driver warning systems. Ninety percent of drivers were in favor of the two-stage system (particularly for fatigue warning) given its potential to reduce the onset of a dangerous event via the cautionary warning and any startle effects that might otherwise arise through use of only a single (urgent) warning.

Warning modalities. Tactile warnings were consistently preferred over visual and auditory warnings as they were considered less likely to be confused with other similar types of stimuli in the cabin. However, drivers indicated advantages in simultaneously presented multiple warnings, noting the use of redundancy to increase the likelihood that at least one warning would be detected if another was missed.

Drivers preferred tactile warnings to be delivered via the seat as opposed to the steering wheel: this view was held both by drivers who had previously experienced steering wheel warnings and those who had not. Warnings delivered through the steering wheel were perceived to be potentially dangerous as they could startle the driver causing them to run off the road; interfere with steering control; or be confused with vibrational feedback from the road during normal driving.

Drivers felt that auditory warnings were more effective than visual warnings at attracting attention as both fatigue and distraction can draw the driver's attention away from the forward roadway where visual warnings may be missed. There was little support for auditory warnings presented as icons (e.g. car horn or screeching tires), with drivers preferring an abstract tone to reduce the likelihood of an icon warning being confused with the sound that it was intended to represent.

4) *Additional Design and Operational Considerations:* A number of other requirements were outlined by drivers, including the need to minimize false positive warnings through better design, including device placement within the cabin, and detection algorithms. Those with experience using driver warning systems felt strongly that the system should not

include functionality that would allow it to be disabled under any circumstances.

D. User Needs and Design Requirements

Building on the findings of the above three stages, a table of 10 user needs and design requirements was created. This is shown in Table II below, in which the first column presents user needs emerging from the literature review, context of use description and truck driver interviews, and the second column shows both how the design would meet these needs and gives supplementary evidence where necessary.

These above 10 points were used as the starting point to develop a basic HMI design concept for discussion and further refinement at two workshops as described below. Participants were presented with an early draft design concept for fatigue and distraction. The fatigue system was a two-stage warning system. This comprised a visual advisory warning (i.e., coffee cup symbol) with a low-level auditory warning presented simultaneously and then a multi-modal urgent warning comprising visual-auditory and tactile alerts. Tactile alerts were emitted via the steering wheel or through the seat. The distraction system was also a two-stage distraction warning system. This comprised a low-level auditory warning (e.g. an auditory ‘earcon’ ping sound) and then a multi-modal urgent warning comprising visual, auditory and tactile alerts.

E. Workshop 1- Drivers/Trainers

Eight participants including four drivers, two driver trainers, one driver operations manager and one general administration manager participated in the first workshop, along with two members of the research team. The approach was to work through the user needs/requirements to help shape the emerging design: as such, a participatory ergonomics approach with end-users [13]. The workshop lasted two hours.

1) *What do you think of a two-stage approach to a warning system?*

All participants were supportive of a two-stage warning system for fatigue and distraction monitoring. Those with experience of single stage warning systems found them to be startling and there was concern that they could be potentially dangerous. Participants agreed that an initial cautionary warning would remove the startling effect by forewarning drivers of an impending urgent warning if their impairment failed to subside or increased. As such, the multi-modal escalating HMIs in the initial design were not only supported by drivers, but actively proposed by them to avoid startle.

Participants also made their own suggestions about potential designs for a two-stage warning system, which were received favorably by the group. The main suggestion was a head-up display (HUD) coupled with an auditory buzzer tone and verbal message.

2) *Display the information effectively.*

With regard to fatigue warnings, most participants suggested that the earcons could be replaced with verbal speech warnings that they felt would be more meaningful and prevent confusion with the many other sounds in the cabin, particularly at the cautionary stage where the warning is less likely to be

TABLE II
USER NEEDS AND DESIGN REQUIREMENTS

User needs. The device should:	How the design would meet these needs:
1. <i>Use a ‘graduated’ approach (warnings hierarchy) for fatigue and distraction warnings</i>	A 2-stage system: advisory information, followed by urgent warning that increases in intensity/salience as the level of fatigue or distraction increases. Evidence/rationale Preferable to have a low-level cautionary alert before the point that more severe indicators of impairment are present so that the driver can take corrective action (e.g., pull off the road and rest) [36,43].
2. <i>Display the information effectively to drivers</i>	The warnings given in the HMI should be slightly different for fatigue and distraction. - For fatigue, a visual advisory warning (e.g. coffee cup symbol) with a low-level auditory warning presented simultaneously, then a multi-modal urgent warning comprising visual-auditory and tactile alerts (via the seat as opposed to steering wheel). - For distraction, a low-level auditory warning (e.g. an auditory ‘ping’ tone) and then a multi-modal urgent warning of visual, auditory and tactile alerts. Evidence/rationale Redundancy overcomes problems with a single mode of warning not capturing the driver's attention (i.e., auditory warning not being heard in a noisy environment, tactile not being felt on a rough roadway) [26-27, 36, 39-40]. Because of their more intrusive nature, the redundancy of the warnings may also more effectively encourage the driver to stop driving in an impaired state [43].
3. <i>Have continuous displays of fatigue/distraction?</i>	No clear information in the literature about whether use of a continuous display of driver state, for example, a gauge showing driver alertness levels, is beneficial, particularly in situations where the driver is not fatigued or distracted. Evidence/rationale Continuous visual feedback was available in some current fatigue warning systems.
4. <i>Require driver input?</i>	No clear information in the literature regarding the potential benefits of driver feedback/ acknowledgement. The danger is that it may increase driver workload or annoyance when acknowledging a warning. Evidence/rationale None of the drivers interviewed thought that driver input was necessary for the warning system to function appropriately. The main reason was that the system should continue to warn the driver if he/she is impaired as the driver is unlikely to be an accurate judge of their own level of impairment.
5. <i>Display the information in a timely manner to drivers</i>	The warnings should be given as soon as possible after the fatigue or distraction event is detected by the system's algorithm in order to trigger an appropriate driver reaction [30-31]. The timing is a balance between being presented early enough so that drivers can successfully correct their behaviour, and not being presented too early so that drivers lose trust in the system and perceive the warnings as annoying or as

heard above other sounds. An example of a cautionary verbal speech warning was ‘you are getting tired’. Likewise, some participants suggested that the visual warning might be better presented in the form of a HUD.

TABLE II
(Continued) USER NEEDS AND DESIGN REQUIREMENTS

	false alarms [51-55]. It is likely that this balance can only be successfully achieved through iterative testing involving drivers.
6. <i>Be tamper proof and not be modifiable.</i>	The warnings should not be disabled / inactivated, except in a limited number of extreme situations. Evidence/rationale All drivers interviewed stated that drivers should not routinely be able to disable the warning system. The primary reason for this view was for the maintenance of safety. However, some drivers were in favour of temporarily disablement, but only if the system emits a high number of false positive warnings. In a small number of currently available fatigue monitoring systems, the alerts could be cancelled and reactivated when the driver pressed a reset button.
7. <i>Be located in a position that does not cause additional distraction or visual demand.</i>	The warnings/HMI should be positioned in an appropriate place. Evidence/rationale Visual displays should be located within the drivers' primary display location: between 5 and 40 degrees below horizontal eye height, and less than 35 degrees laterally [47].
8. <i>Not interfere with other in-cab systems.</i>	The system should not interfere physically or cognitively with the task of safe driving [47-50].
9. <i>Not require extensive training to understand and use the system.</i>	The HMI design should not require extensive training to understand and use the system [47, 49-50]. However, some degree of training may assist with improving system effectiveness and acceptance [23, 56].
10. <i>Be trusted and accepted by drivers.</i>	The system should be actively supported by company management [1, 5]. The design of the HMI should be user-centred (through driver interviews and workshops) to increase trust and acceptance [7, 56]. The system should avoid false alarms (false positives) as much as possible [36, 51-55]. This may require user testing and feedback to define what is an acceptable false alarm rate.

Participants generally felt that tactile warnings issued at the cautionary warning stage would be unnecessary, although some suggested that an intermittent pulse for the cautionary warning and a constant pulse for the urgent warning could overcome the risk that drivers would not detect the first warning. A tactile warning emitted through the seat was perceived to be better than through the steering wheel as most drivers thought the latter could produce a startle response which may be dangerous.

Participants had similar views towards the distraction warning system as for the fatigue warning system. Although all drivers thought there should be a different type of warning for fatigue and distraction, there was a general consensus that the same design concept with cautionary and urgent warning stages should apply for both states, partly to avoid increasing the number of different warnings in the vehicle's cabin.

3) What do you think of continuous displays of alertness?

Participants were generally not opposed to the concept of a display showing the driver's alertness level continuously but most were concerned about how this would be integrated into

the cabin which they felt was already very cluttered. Some participants thought there was no need for such a system if the fatigue and distraction warning system was two level. The general perception was that drivers need only know when they are fatigued/distracted or approaching such a state, and not when they are alert.

4) Does there need to be some acknowledgement of the system?

Some participants suggested that driver acknowledgement of the system might be useful if it helped to keep them awake. However, most felt that there would be no benefit to having it as it may cause annoyance or distraction, especially in the case of false alarms.

5) Display the information in a timely manner to drivers

Participants stated that it was more important that the system be accurate over fast and that they would be reluctant to accept too many false alarms. It was understood by all that balancing accuracy with timeliness here may be best achieved through future iterative testing involving drivers.

6) Tamper proof?

All participants agreed that the system should be tamper proof and should never be able to be turned off or include functionality for its settings to be modified (e.g., volume of warnings turned down or up).

Other comments

All participants agreed that the safety benefits of a driver fatigue and distracting warning system would likely extend beyond the initial short-term effect of alerting the driver to their impaired state. Potential long-term benefits suggested by drivers included helping them to identify triggers for fatigue or distraction before the system emits a warning, and to better plan and manage their shifts for improved overall health and well-being. Those with previous experience of driver monitoring reported that their initial perceptions of the technology as an invasion of privacy had changed over time and they were now more willing to view it as a means of protecting their safety.

F. Workshop 2- Managers/Developers

The second two-hour design workshop had nine participants who were managers and senior supervisors from the trucking company. The approach presented the revised design and sought feedback, especially regarding how the company may use the information provided by such a system.

1. What do you think of the two-stage design for fatigue and distraction warning?

Distraction warning: Some participants suggested that a single, continuous warning that potentially escalates after a given period might be better than a two-stage warning system for distraction. Unlike fatigue where a driver can cycle between fatigued and alert states without awareness, managers felt that drivers have more control over engaging in distracting activities (i.e., they can start and stop whenever they so choose) and the task tends to be more continuous unless the driver deems it unsafe to continue. This point was accepted by all participants.

Fatigue warning: The two-stage design was accepted for fatigue. Regarding auditory warnings, as for the distraction

warning, drivers preferred use of an abstract alert tone over a representational alert tone (such as a car horn or screeching tires) as it may be confused with the tones that they are intended to represent (i.e. a real car horn or real screeching tires). Combining an alerting tone with a short verbal message was viewed positively.

2. *What does the company want the driver to do in response to a first stage warning, particularly for fatigue?*

Almost all participants wanted drivers to self-manage their fatigue or distraction at the first level of warning (i.e., level 1 warning stage) rather than receive a telephone call from the company to check on their status. However, there was strong agreement that the level 1 warnings should be logged by the company as a means of monitoring and managing drivers, particularly those who regularly received such warnings.

3. *If the driver is to self-manage at the level 1 warning stage, then what does the company want them to do, specifically?*

Most participants suggested that the driver's response to a level 1 warning should depend on the driver's individual needs. A one size fits all approach was deemed to be ineffective even though it was acknowledged that general guidelines as to what works best to boost safety would be given during training (e.g., taking a break when tired).

One of the managers posed the question of whether it is important to know what the driver actually does if they stop to take a break after receiving the level 1 warning. Most participants agreed that this type of information could prove useful and could be easily collected by having regular meetings with drivers who had been logged by the system as having received multiple level 1 warnings.

G. Operationalizing the HMI Concepts

The HMI concepts developed through the above process were then 'operationalized': that is, turned into specific design criteria for engineers to install into a truck simulator prior to evaluation. This process was primarily based on the HMI guidelines and literature previously reviewed; where specific information was not available then input was sought from subject matter experts (that is truck drivers) and ergonomics 'rules of thumb' were applied or professional engineering judgments were made in conjunction with the manufacturers. To demonstrate this process, abridged results of the initial operationalization for the distraction HMI design concept are shown in Table III.

H. HMI Evaluation

The evaluation of the developed HMI designs used two methods: user-based testing and inspection-based. Both approaches are recommended in ISO 9241-210 [8]. A summary of the methodology and outcomes is given below.

In the user-based evaluation, each participant was tested individually: 14 truck and car drivers (11 male, 3 female; mean age 38.4 years; mean driving experience nearly 18 years). Participants arrived at the simulator facility, provided informed consent and completed a demographics and driving experience questionnaire. They were given an overview of the study. The participant was accompanied to the truck simulator and set up

TABLE III
INITIAL OPERATIONALIZED SPECIFICATIONS FOR THE DISTRACTION HMI

Non-escalated	Specifications
Audio	<i>Alerting tone ('beep' immediately after the system's detection algorithm has indicated the driver is distracted)</i> • Complex tone [44] • Harsh soundwave [44] • Frequency: 1000Hz [44-45] • Two pulses [36,44] • Pulse duration: 200 ms [36, 43-44, 46] • Inter-pulse interval: 150 ms [44-45] • Volume: 70 DbA [47, 22] • Onset rate between 1 dB/msec and 10 dB/msec [44]. <i>Speech message</i> • Starts 500 ms after beep [44-45] • 'Pay Attention' [43, 57] • Spoken in authoritative [43] • Australian (local) female voice [58]. • Volume: 70 dBA [47, 57].
Tactile	• Five individual pulses via the vibratory motor used in current Guardian System (Seeing Machines) tactile warning • Pulse duration: 200 ms [43] • Inter-pulse interval: 200 ms [43] • Vibrational frequency 100 to 300 Hz [43] • Duty cycle: 0.3 (ensures lower 'intensity' relative to Escalated) • Intensity: 20 to 30 dB above masked vibratory threshold [43] • Triggered at the same time as the audio alerts [36, 44, 47].
Escalated	
Audio	<i>Alerting tone ('Beep')</i> As for non-escalated, except • Frequency: 1500 Hz [44-45] • Four pulses [44] • Pulse duration: 105 ms [36, 43-44, 46] • Inter-pulse interval: 30 ms [36, 44] • Volume: 90 dBA [36, 47, 57]. <i>Speech message</i> • Starts 350 ms after beep[44-45] • 'Pay Attention now!' [47, 57] • Volume: 90 dBA [47, 57]
Visual	• 'Eye' graphic in red (14, 20) against road displayed in white. • Presented as HUD • Not in driver's central field of view [36, 47], but located within +/- 15 degrees of the driver's normal line of sight [36, 44] • Graphic flashes for 300 ms (on), followed by 100 ms off. Repeated 5 times (total of 2000 ms). Following this the graphic remains on for 1000 ms [43-44, 47] • Triggered at the same time as the audio alerts [36, 44, 47]
Tactile	As for non-escalated, except duty cycle: 1.0 (to ensure higher 'intensity' relative to Non-escalated).

in the cab. For the simulator trial, firstly a five-minute familiarization drive was given. When participants were comfortable with the simulator, the trial began and the warnings were presented to drivers at predefined intervals. The evaluation did not measure driver behavior per se, but instead focused on warning comprehension, acceptance, perceived effectiveness and driver opinions.

The user-based results found that HMIs were accepted and considered useful by the user-based participant group

that included both experienced truck and car drivers. The different warning elements in the HMIs were noticed, correctly understood and were considered effective by the participants. Equally, they were perceived as not likely to excessively interfere with other in-cab warnings and not overly likely to overload or startle a driver.

In the inspection-based review, the HMIs were also audited against best practice by means of two checklists/usability principles including the Design Guidelines for Safety of In-Vehicle Information Systems produced by the Transport Research Laboratory in the UK [47]. They revealed no major design deficiencies. The recommended design iterations from both type of evaluations were:

- Include a system status button.
- Be careful about allowing adjustment in either auditory or visual warnings; potentially just include a day/night mode for the visual warning.
- Increase the volume of the auditory tone and voice, as it was not always perceived with background noise.
- Driver training/information by a company prior to use.
- Increase the escalation period in the distraction warning to 2 seconds. The inspection based review noted 1.5 seconds “...may not provide enough time for distracted drivers to reorient their attention back to the roadway”.
- More broadly, manage the process of technology deployment. Ideally management should openly explain the benefits of such systems and describe how the distraction and fatigue information would be used. Allowing for ongoing driver feedback about the system is also important, especially for issues such as false alarms and driver acceptance of the system.

I. Design Finalization

The design concepts went through several iterations during the entire HCD process. For both fatigue and distraction these changes included: providing an initial alerting tone for the auditory warning (and recommended by the workshops), having a spoken message auditory warning (rather than an abstract sound), having the distraction system as escalating (rather than two separate levels, as for fatigue), modifying the tactile warning for the cautionary level, and using a HUD for the visual warning (and recommended by the driver workshop). Similarly, the evaluations for the distraction concept both recommended changing an escalation time of 2 seconds, the addition of a ‘system status button’ and increasing the volume of the auditory tone and voice.

Based on the HCD processes and evaluations described above, a summary of the final HMI concepts for both distraction and fatigue are presented below.

1) Distraction Warning System- HMI Design Concept: The distraction warning system is an escalating HMI with cautionary then urgent warnings. Following the cautionary warning immediately after the system’s detection algorithm has indicated the driver is distracted, the system would escalate to the urgent warning level after two seconds if the driver is still displaying distracted behavior. If the driver stops displaying distracted behavior then the system would not progress



Fig. 2. Visual warning ‘eye & road’ as a HUD on the windscreen inside a truck simulator cab.

to the escalated level. The system would give auditory, visual and tactile (seat) warnings. In summary:

a) Caution Level:

- Auditory warning: alerting tone then simple spoken message in a local female voice (‘pay attention’)
- Visual warning: nothing presented
- Tactile warning: intermittent pulsing of seat.

b) Escalated — Urgent-Level (After Two Seconds, if the Driver is Still Displaying Distracted Behavior):

- Auditory warning: louder, more urgent alerting tone then simple spoken message in a local (Australian) female voice (‘pay attention now!’)
- Visual warning: eye graphic as a HUD on the windscreen (Fig 2).
- Tactile warning: full seat vibration (this is the same as for the fatigue tactile warning).

2) Fatigue Warning System- HMI Design Concept: The fatigue warning system is two-level HMI with cautionary then urgent warnings. The system will give auditory, visual and tactile (seat) warnings.

a) Level 1 (Caution):

- Auditory warning: alerting tone then simple spoken message in a female voice (‘caution fatigue’)
- Visual warning: amber coffee cup as a HUD on windscreen
- Tactile warning: intermittent pulsing of seat.

b) Level 2 (Urgent):

- Auditory warning: louder, more urgent alerting tone then spoken message in a female voice (‘danger fatigue!’)
- Visual warning: larger, flashing red coffee cup as a HUD on windscreen
- Tactile warning: full seat vibration (as for the distraction tactile warning).

IV. DISCUSSION

This case study outlined the employment of a human-centered design process for the HMI of an in-vehicle driver warning system. A multi-stage process was used with the principles of human-centered design at its heart, building on leading HCD design methods employed in similar domains [7] and participatory ergonomics programs [13].

Through this HCD process, core design specifications for driver fatigue and driver distraction systems were developed

and then tested against an inspection-based evaluation and among a group of car and truck drivers (user-based evaluation). This work therefore developed and applied a HCD-focused design process to driver warning technology, and the subsequent HMI was a) perceived by drivers to be efficacious, and b) had a high level of end-user acceptability and usability. The novel contribution of this research therefore was using a HCD process as defined in ISO 9241:210 to develop an effective HMI for fatigue and distraction for commercial vehicles.

A key advantage of this HCD approach was the inclusion of end-user feedback through interviews, design workshops and user-based evaluation. This participatory approach was particularly beneficial for the HMI design concept iteration given the knowledge gaps identified in the literature review on driver state monitoring HMI for fatigue and distraction. The drivers, trainers and managers employed here have extensive experience in commercial vehicle operations: for example, 75% of the initial truck drivers interviewed had direct experience with currently available driver warning systems for over 18 months.

As noted in the introduction, although having a well-designed warning system is necessary, the system should ideally be integrated as part of a commercial vehicle operator's fatigue management plan [5]. So, if a fatigue and distraction warning is issued by the system then it is not treated as an isolated event but instead part of a package that includes training and a joint management/driver review of such events [6]: it is anticipated here that such a learning loop might also involve system developers to help further refine the warnings given.

One particular challenge in this HCD work was to reconcile the 'naïve' perspectives of the drivers (and other end-users such as trainers and supervisors) with the theoretical understanding developed from the scientific literature. For example, the interview section of this HCD process highlighted how the drivers exhibited a preference for single mode (auditory) warnings, whereas, as seen in the literature review, the science demonstrated the vulnerability of auditory-only warnings to remain 'unheard'. Thus the final design included multi-modal warnings, reconciling end-user/driver wishes with scientific evidence. It is therefore important to emphasize that HCD with a participatory focus is not simply designing according to the wishes of the end-users, but rather an active process of best-reconciliation between end-users' desires and existing scientific evidence.

V. CONCLUSION AND IMPLICATIONS

This research provided an opportunity to enhance driver warning systems through developing and evaluating an HMI by means of an iterative HCD process. The outcomes of this research underscore the importance of using an iterative approach as part of the HCD process. From this process, fit-for-purpose design concepts were developed, thus meeting the initial aims of developing an effective HMI using a HCD process. Whether the design concepts developed here translate to real-world improvements in driver safety by means of accurate and robust distraction and fatigue warnings requires a

comprehensive on-road evaluation across a range of commercial vehicle operations. This evaluation also requires that the fatigue and distraction warning are integrated and harmonized with other in-cabin warning systems such as lane departure warnings; this will also confirm that no additional distraction is produced by the new HMI. Such on-road testing is recommended to be undertaken in the near future.

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